Second Law of Electrolysis

It states that the masses of the resulting separated elements are directly proportional to the atomic masses of the elements, when an appropriate integral divisor is applied.

$$W \propto E \qquad or \qquad \frac{W_1}{W_2} = \frac{E_1}{E_2}$$

Electrochemical Series

When metals are arranged in the increasing order of their standard reduction electrode potential, a series is obtained, which is called electrochemical series. It is also called **reactivity series** (as reactivity follows the reverse order). Lower is the value of reduction potential, greater would be its reducing power.

Depending upon the reactivity, the occurrence of metals in the reactivity series may be represented as shown in the figure.

Metals above hydrogen:
- K, Ca, Na, Mg, Al — Highly reactive metals never found in free state
- Zn, Cr, Fe, Ni, Sn, Pb — Moderately reactive metals

H

Metals below hydrogen:
- Cu, Hg, Ag — Less reactive metals
- Pd, Pt, Au — Least reactive metals (found in the free state)

Thus, gold is the least reactive metal and hence, found in native state. The more reactive metals of the series can displace the less reactive metals from their salt solutions.

e.g. $Cu + FeSO_4 \longrightarrow$ No reaction

but $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$

Moreover, metal placed before H can evolve H_2 from dilute acids.

e.g. $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$

The metals placed above are good reducing agent and reduces the metal oxides of the metal placed below them.

CORROSION

It is the process of oxidative deterioration of a metal surface by the action of environment to form unwanted corrosive products. e.g. conversion of iron into rust ($Fe_2O_3 \cdot xH_2O$). Corrosion of iron is called **rusting**.

Tarnishing of silver (due to the formation of Ag_2S), development of green coating [of $Cu(OH)_2 \cdot CuCO_3$ (basic copper carbonate)] on copper and bronze. It is basically an electrochemical process.

It is accelerated by the presence of impurities, H^+, electrolytes such as NaCl and gases such as CO_2, SO_2, NO, NO_2 etc.

Prevention of Corrosion

- By coating with a suitable material (barrier protection *viz* paint, grease, metals (like Zn, Cu, Ni, Cr, Sn), adherent film of iron phosphate, adherent coating of magnetic oxide (Fe_3O_4).
- By alloying with suitable metals.
- By cathodic protection–the metal is made cathode by application of external circuit.
- By using artificial anode (sacrificial protection), more anodic metal (like Zn, Mg, Al etc.,) is connected to iron article by a wire. In such cases, current flows towards more reactive metal and prevents iron from corrosion.
- By galvanisation (covering of iron article with zinc).

INORGANIC CHEMISTRY

CLASSIFICATION OF ELEMENTS

- In 1896, Mendeleev gave a periodic law. According to which "the properties of elements are the periodic function of their atomic masses".
- There were seven periods (horizontal rows) and eight groups (vertical columns) in the periodic table of Mendeleev having 63 known elements at that time.
- Moseley modified Mendeleev's periodic law and proposed modern periodic law. According to **modern periodic law**, "the properties of elements are periodic function of their atomic numbers."
- There are eighteen vertical columns, known as **groups** and seven horizontal rows, known as **periods**, in the long form of periodic table.
- **Periodic Properties** The properties which are repeated at regular intervals are known as periodic properties. Atomic and ionic size, electron affinity, ionisation potential, electronegativity, electropositive character, acidic or basic character, metallic nature, etc., are some important periodic properties.

Characteristics of Periods

- The number of valence electrons in elements increases from 1 to 8 on moving from left to right in a period.
- The elements in a period have consecutive atomic numbers.
- The valency of the element increases from 1 to 4 and then decreases to 0 (zero) on moving from left to right in a period, with respect to hydrogen.
- Atomic size, electropositive character, metallic character, reducing nature of elements and basic nature of oxides all decrease from left to right in a period.

- Electronegative nature, non-metallic nature, acidic nature of oxides, ionisation potential increases from left to right in a period. In a period, electronegativity and electron affinity also increases from left to right.

Characteristics of Groups

- All the elements of a group of the periodic table have the same number of valence electrons and hence, have almost similar chemical properties.
- Atomic radius, electropositive nature, metallic nature, reducing nature of elements and the basic nature of oxides increase from top to bottom in a group.
- Electronegative nature, ionisation potential, electron affinity, electronegativity, non-metallic nature and acidic nature of oxides decrease down a group with increasing atomic number.

HYDROGEN

- Hydrogen is the first element in the periodic table with atomic number 1 and mass number 1.
- Hydrogen has three isotopes-protium or ordinary hydrogen $({}_1H^1)$, deuterium or heavy hydrogen (${}_1H^2$ or ${}_1D^2$), tritium (${}_1H^3$ or ${}_1T^3$). Tritium is the radioactive isotope of hydrogen.
- Hydrogen is a non-supporter of combustion and burns with 'pop' sound. It is an inflammable gas.

Compounds of Hydrogen

Water is one of the most important compound of hydrogen.

Water

- Water is the neutral oxide of hydrogen. Pure water freezes at 0° C and boils at 100° C. This abnormal high boiling point is due to the association of H_2O molecules through hydrogen bonding.
- Water is a polar compound (dipole moment 1.85 D) and possesses a high dielectric constant, i.e. 81. Its presence is tested by using anhydrous copper sulphate which turns blue in the presence of water.
- Water is a universal solvent because of its tendency to form H-bond and polar nature.
- Water has very high specific heat and thus, widely used as a coolant.
- Water which form lather with soap is called **soft** and which does not form lather is called **hard** water as the ions present in it form scum (precipitate) with the soap.

Hardness of Water

- Hardness of water may be due to the presence of bicarbonates of calcium or magnesium (**temporary hardness**) or due to the presence of chlorides or sulphates of Ca or Mg (**permanent hardness**).

Removal of Hardness of Water

- Temporary hardness is removed by boiling or by adding lime water (Clark's process).
- Permanent hardness is removed by adding sodium carbonate (washing soda,Na_2CO_3) or calgon (sodium hexa metaphosphate, $Na_6P_6O_{18}$ or zeolite which is also called permutit (hydrated sodium aluminium silicate, $Na_2Al_2Si_2O_8 \cdot xH_2O$).
- From sea water, pure water is obtained by a process, called **reverse osmosis**, i.e. by applying pressure higher than the osmotic pressure towards the solution side. This process is also called desalination of sea water.
- Boiling, chlorination and treatment with UV irradiation all kill the microorganisms present in water.

Heavy Water

- **Heavy water** (D_2O, molecular weight = 20) is the oxide of heavy hydrogen. It is used as a moderator in nuclear reactors.
- In hydrogen peroxide (H_2O_2) oxygen is in −1 oxidation state, which is the intermediate oxidation state of oxygen. H_2O_2 behaves as an oxidising, reducing and bleaching agent.
- Lead painting becomes blacken in air due to the formation of PbS by the action of H_2S present in air, H_2O_2 is used to wash it. H_2O_2 oxidises PbS to $PbSO_4$.

Aquifers

An aquifer is an underground layer of water bearing permeable rock, i.e. they have opening through which only liquids and gases can pass.

Clay layers although are poor aquifers. This is because clay minerals are dense impermeable materials. They act as aquiferal, i.e. a layer of material that is almost impenetrable to water.

Metallurgy

It is the process of extraction of metals from their ores is called **metallurgy**.

Terminology Related with Metallurgy

- ***Metals*** occur in the form of minerals in nature.
- ***Gangue*** or matrix are the impurities associated with ore.
- ***Roasting*** is the process of heating ore (mainly sulphide ore) in the excess of air whereas calcination is heating the ore (carbonate or hydroxide ore) in the absence or limited supply of air.
- ***Flux*** is added to gangue to convert it into slag.
- ***Ores*** are the minerals from which metal is extracted conveniently and beneficially.

Some Important Ores and their Uses

Metal	Ores/minerals	Chemical composition
Sodium	Rock salt Chile salt petre Borax, tincal or suhaga carnallite	$NaCl$ $NaNO_3$ $Na_2B_4O_7 \cdot 10H_2O$
Magnesium	Magnesite Asbestos Carnallite	$KCl, MgCl_2, 6\ H_2O, MgCO_3$ $CaSiO_3 \cdot 3MgSiO_3$ $KCl \cdot MgCl_2 \cdot 6H_2O$
Calcium	Lime stone Gypsum Fluorspar	$CaCO_3$ $CaSO_4 \cdot 2H_2O$ CaF_2
Aluminium	Bauxite Cryolite Feldspar Mica Granite	$Al_2O_3 \cdot 2H_2O$ Na_3AlF_6 $KAlSi_3O_8$ $KAlSi_2O_{10}(OH)_2$ SiO_2 and Al_2O_3 (73 : 14)
Iron	Haematite Magnetite Iron pyrites Siderite	Fe_2O_3 Fe_3O_4 FeS_2 $FeCO_3$
Copper	Copper glance Copper pyrites Malachite Azurite	Cu_2S $CuFeS_2$ $Cu(OH)_2 \cdot CuCO_3$ $2CuCO_3 \cdot Cu(OH)_2$
Silver	Silver glance Horn silver Ruby silver	Ag_2S $AgCl$ $Ag_2S \cdot Sb_2S_3$
Gold	Sylvanite	$AuAgTe_4$
Zinc	Zinc blende Calamine Zincite	ZnS $ZnCO_3$ ZnO
Mercury	Cinnabar	HgS
Tin	Cassiterite	SnO_2
Lead	Galena Cerrusite Anglesite	PbS $PbCO_3$ $PbSO_4$
Potassium	Nitre Sylvine	KNO_3 KCl

Alloy

- An alloy is a homogeneous mixture of two or more metals or a metal and a non-metal :

Some Important Alloys and their Uses

Alloys	Composition	Important uses
Solder	Tin and lead	Soldering
Bronze	Copper and tin	Making utensils, statues, coins, etc.
Type metal	Tin, lead and antimony	Used in printing
Bell metal	Copper, tin	Making bells
Gun metal	Copper, tin and zinc	Gears and bearing
Brass	Copper, zinc	Utensils, condenser, tubes, cartridge caps, etc.
Aluminium bronze	Copper and aluminium	Coins, picture, cheap jewellery, flames
German silver	Copper, zinc, nickel	Utensils, resistance wires
Constantan	Copper, nickel	Electrical apparatus
Dental alloy	Silver, mercury, tin, zinc	For filling teeth
Stainless steel	Iron, chromium. nickel	Utensils, bicycle parts, etc.
Magnalium	Magnesium and aluminium	Automobile and aeroplane parts
Nichrome	Nickel, iron, chromium, manganese	In making coils of heater
Misch metal	Cerium, lanthanum, neodymium, praseodymium and other lanthanoids	In making cigarette lighters

Elements of IA Group (Alkali Metals)

- Lithium, sodium, potassium, rubidium and caesium are alkali metals and are *s*-block elements.
- These metals are soft and can be cut with knife.
- Lithium is the lightest metal.
- Alkali metals are stored under kerosene or paraffins to protect them from the action of air.
- Lithium shows diagonal relationship with magnesium.
- In Castner's process, metallic sodium is prepared by electrolysis of molten NaOH.
- Sodium is used in yellow light lamps.
- Sodium chloride (NaCl) or common salt or table salt is used in our daily diet, as a preservative for pickles, meat and fish. It is also used in the manufacture of NaOH, Cl_2 gas and soap.
- Sodium hydroxide (NaOH) or caustic soda is used in the soap, dyes and artificial silk industries and in the refining of bauxite mineral.
- Sodium bicarbonate is used in the kitchen for making food tasty and crispy.
  - Sodium bicarbonate (Sodium hydrogen carbonate or baking soda) is used for making baking powder, which is a mixture of baking soda and a mild edible acid such as tartaric acid.
  - It is also an ingredient of antacids. Being alkaline, it neutralises excess acid in the stomach and provides relief.
  - It is also used in soda acid fire extinguishers.
- Sodium carbonate decahydrate ($Na_2CO_3 \cdot 10\ H_2O$) or washing soda is used in the manufacture of glass, soap, washing powder and for softening hard water.
- Sodium sulphate ($Na_2SO_4 \cdot 10H_2O$) is Glauber's salt. It is used as purgative.
- Sodium thiosulphate ($Na_2S_2O_3 \cdot 5H_2O$) is also known as hypo. It is used in photography as a fixing agent because it removes the undecomposed AgBr as soluble silver thiosulphate salt.

Elements of IB Group

- $_{29}Cu$, $_{47}Ag$, $_{79}Au$ are the elements of **I B group**.
  These elements are called coinage or currency metals. Their important properties are as follow:
  - They are *d*-block, transition elements.
  - These are very hard, malleable and ductile metals.
  - They have high density and high melting and boiling points.
- $CuSO_4$ is used as a mordant in dyeing. Copper sulphate pentahydrate is blue in colour and is called **blue vitriol**. $CuSO_4$ is used to test the presence of water.
- Copper is alloyed with gold and silver for making ornaments.
- $AgNO_3$ is called **lunar caustic**. It is stored in dark brown bottles because it is photosensitive.
- Au, Ag, Pt, etc., metals which remain unaffected by the action of other strong acids dissolve in aqua-regia (which is a mixture of HCl and HNO_3 in 3 : 1).

Elements of IIA Group

(Alkaline Earth Metals)

- II A subgroup elements are Be, Mg, Ca, Sr, Ba and Ra. They are called **alkaline earth metals** and are *s*-block elements.
- Calcium of this family is most abundant element in the earth's crust while magnesium is present in chlorophyll, the green pigment of the leaf.
- Beryllium shows diagonal relationship with aluminium.
- $Mg(OH)_2$ is called **milk of magnesia**. Being a mild base, it is used as an antacid to neutralise the excess acid present in the stomach.
- $MgSO_4$ is used as a mordant in dyeing and tanning industry. It is also used as a purgative.
- Calcium oxide (CaO), also called quicklime, gives hissing sound when dissolved in water. It is used in the manufacture of glass, calcium chloride, cement, mortar, bleaching powder, calcium carbide, slaked lime, etc. It is also used in the extraction of iron and as a drying agent for ammonia and alcohol.
- Calcium hydroxide (slaked lime) $[Ca(OH)_2]$, is used in the manufacturing of bleaching powder, caustic soda and soda lime and for softening of hard water.
- When $CaCO_3$ is dipped in water, the bubbles evolve due to the evolution of carbon dioxide.
- Calcium sulphate is gypsum $(CaSO_4 \cdot 2H_2O)$. It loses a $\left(\frac{2}{3}\right)$rd part of its water of crystallisation when heated at 120°C to form $(CaSO_4)_2 \cdot H_2O$ which is known as Plaster of Paris.
- Gypsum is used in the preparation of building's plaster and anhydrous, $CaSO_4$.
- Anhydrous $CaSO_4$ is a dehydrating agent and is used for the manufacture of ammonium sulphate (Sindri fertiliser) and sulphuric acid.
- Plaster of Paris is a white powder, which sets into hard mass called gypsum on hydration with water, so it is used in making casting for statues, toys, for plastering fractured bones, etc.
- Cement has the following composition: Calcium oxide (CaO, 50-60%), alumina (Al_2O_3, 5-10%) and magnesium oxide (MgO, 2-3%). Raw materials used for its preparation are limestone and clay. When cement mixed with water, the setting of cement takes place to give a hard mass.
  This is due to the hydration of the molecules of the constituents and their rearrangement.
- Adding gypsum is only to slow down the process of setting time of the cement.
- Portland cement is a mixture of silicates and aluminates of calcium with small amount of gypsum. The chemical composition of portland are CaO, SiO_2, Al_2O_3, $SO_3 MgO, Fe_2O_3$, silica SiO_2 imparts strength to the cement due to the formation of dicalcium and tricalcium silicates.
- Alumina (Al_2O_3) imparts quick setting property to the cement. It acts as a flux and lowers the clinkering temperature.

Elements of IIB Group

- Zinc ($_{30}Zn$), cadmium ($_{48}Cd$) and mercury ($_{80}Hg$) are the three elements present in II B subgroup. These are *d*-block elements.
- Their properties are as follow:
  - Zinc is used in making alloys like brass, bronze, German silver, etc.
  - Zinc is deposited on the surface of iron articles by the process called galvanisation.
  - Mercury, Hg is a liquid at room temperature and is filled in CFL lamps.
  - Mercurous chloride or calomel(Hg_2Cl_2) is used for making calomel electrode which is a reference electrode.

Elements of III A Group (Boron Family)

- Elements are $_5B$, $_{13}Al$, $_{31}Ga$, $_{49}In$ and $_{81}TI$.
- Boron is a semi-metal. Rest of elements are metals.
- Aluminium is the most abundant metal (and third most adundant element) in the earth's crust.
- Boron halides are Lewis acids. The order of acidity is $BBr_3 > BCl_3 > BF_3$.
- Being lighter, all alloys are used in making aeroplanes.
- Aluminium sulphate is used as a mordant in dyeing.

Elements of IVA Group (Carbon Family)

- The IVA group consists of elements C, Si, Ge, Sn, Pb.
- All are *p*-block elements.
- The stability and volatility of tetrahalide of carbon decrease with increasing molecular weight of the tetrahalide e.g. $CF_4 > CCl_4 > CBr_4 > CI_4$.

Carbon

This is the element which forms highest number of compounds (except hydrogen) due to its catenation power (tendency of self linking) and tetravalency. It has two crystalline allotropes : Diamond and graphite.

Diamond

- It is the purest form of carbon and has rigid three dimensional network structure in which each carbon atom is bonded to four other carbon atoms in tetrahedral fashion.
- It is hardest substance and is a bad conductor of electricity.
- It is used in making jewellery.
- It is used as an abbrasive for sharpening hard tools, i.e. in making rock drilling machines , glass cutting etc.

Graphite

- It has layered two dimensional hexagonal structure.
- Large amounts of graphite are prepared artificially by Acheson process.
- Graphite is a good conductor of electricity.
- It is used as dry lubricant and in nuclear reactor as a moderator.

➨ **Note** Fullerenes form another class of carbon allotropes. The first one to be identified was C-60 which has carbon atoms arranged in the shape of football.

Coal and Charcoal

- Coal is the amorphous variety of carbon. It is used as fuel. Coals are of following types :
  - (a) Peat (60% carbon).
  - (b) Lignite or brown coal (70% carbon).
  - (c) Bituminous (78-83% carbon). It is the most common variety.
  - (d) Anthracite (90% carbon).
- Charcoal can be animal charcoal, wood charcoal, sugar charcoal and activated charcoal.
- It is used in the manufacture of fuel gases like-water gas, producer gas and semi-water gas.
- Water gas ($CO + H_2$), semiwater gas (water gas + producer gas), producer gas (N_2 + CO) are obtained from CO.
- Charcoal (animal charcoal) is used as a fuel and also as a deodorant in the purification of water, decolourising sugar solution and in gas masks.

Carbon Dioxide (CO_2)

- Air contains CO_2 to an extent of 0.03%-0.05% by the volume. CO_2 is an acidic oxide of carbon.
- Solid CO_2 is technically known as dry ice (dricold).
- It produces very low temperature (– 100°C) with ether, hence gives an excellent freezing mixture.
- It is a non-supporter of combustion so generally used as fire extinguisher.
- When carbon dioxide is passed through lime water, it turns milky due to the formation of calcium carbonate (which is milky white in colour). However, in excess of the gas, the precipitate dissolves because of the formation of soluble calcium bicarbonate.

$$CO_2 + \underset{\text{Lime water}}{Ca(OH)_2} \longrightarrow \underset{\text{White}}{CaCO_3} + H_2O$$

$$CaCO_3 + \underset{\text{Excess}}{CO_2} + H_2O \longrightarrow \underset{\text{Soluble}}{Ca(HCO_3)_2}$$

- **Carbon dioxide** is the product of calcination and respiration processes.

Silicon

- It is the second most abundant element in the earth's crust.
- Crystalline form of SiO_2 (silica) is quartz. SiO_2 is the major component of glass and is soluble in HF. That's why HF cannot be stored in glass bottles.

Glass

Glass is an amorphous and transparent solid, obtained by solidification of various silicates and borates.

- Various varieties of glass are : Coloured glass, hard glass, high refractive index glass, pyrex glass, Crook's glass.

Types of Glasses and their Uses

Types of Glasses	*Uses*
Soft glass	Window panes, bottles, test tubes, etc.
Hard glass	Combustion tubes, etc.
Crook's glass	Absorbs UV rays, used for making lenses
Pyrex glass	Cooking utensils
High refractive index glass	Lenses, cut glasses
Glass laminates	Bullet proof materials

- Coloured glass can be obtained by adding following compounds.

Compound added	*Colour imparted*
Cobalt oxide	Blue
Cuprous oxide	Red
Cadmium sulphide	Lemon yellow
Chromium oxide	Green
Auric chloride	Ruby
Manganese dioxide	Purple